



Koneru Lakshmaiah Education Foundation

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DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE: ENGINEERING CAPSTONE PROJECT – 1(23IE4053R/23IE4053A)
A.Y. 2026 - 2027

TITLE OF THE PROJECT	RetailIQ – Full-Cycle E-Commerce Customer & Sales Analytics	
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Abstract

RetailIQ is an end-to-end data science and analytics project developed to analyze e-commerce transaction data and transform raw business data into meaningful and actionable insights. The project focuses on understanding sales performance, customer purchasing behavior, product performance, geographical sales distribution, customer segmentation, and future revenue trends. The primary objective of RetailIQ is to provide a comprehensive analytical solution that can support data-driven decision-making in an e-commerce business environment.

The project begins with the collection and understanding of an online retail transaction dataset containing information such as invoice numbers, product codes, product descriptions, quantities, unit prices, invoice dates, customer identifiers, and countries. The raw data undergoes a systematic preprocessing and cleaning process. This includes identifying and handling missing values, removing duplicate records, handling returned products and negative quantities, and eliminating invalid or non-positive prices. After cleaning, feature engineering is performed to create useful analytical attributes, particularly transaction-level revenue, which is calculated using quantity and unit price. Additional date-related features such as year, month, day, and hour are also extracted to support time-based analysis.

Exploratory data analysis is then performed to understand overall sales patterns and business performance. The project analyzes total revenue, number of orders, customer activity, product performance, monthly sales trends, and country-wise revenue. Product-level analysis is used to identify the highest-performing products based on revenue and sales activity, while country-level analysis provides insights into the geographical distribution of sales.

A major component of RetailIQ is customer analytics and **RFM (Recency, Frequency, Monetary) analysis**. Recency measures how recently a customer made a purchase, Frequency

measures the number of orders placed, and Monetary value represents the customer's total spending. Based on these measures, customers are assigned RFM scores and classified into meaningful business segments such as Champions, Loyal Customers, New Customers, Regular Customers, At Risk, and Lost Customers. This segmentation helps identify valuable customers and customers who may require targeted retention strategies.

The project also incorporates revenue forecasting to estimate future sales based on historical monthly revenue patterns. A machine learning-based forecasting approach is used to generate future revenue predictions, providing an additional perspective for sales planning and business decision-making.

Finally, all major analytical results are integrated into an interactive **Streamlit dashboard**. The dashboard presents business KPIs, monthly revenue trends, top products, country-wise sales, customer analytics, RFM segmentation, and revenue forecasts through an easy-to-understand visual interface. Thus, RetailIQ demonstrates a complete data science workflow, from data preprocessing and exploratory analysis to customer segmentation, forecasting, and interactive visualization, providing a practical analytical solution for e-commerce business intelligence and decision support.